General Science

Classes 9-10 Combined - Solved Exercises - PCTB

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Chapter 1: Introduction and Role of Science

Exercise 1 - Q1 Fill in the blanks

Q1 (i) Jabir Bin Hayyan was an expert in _____.

Chemistry

Q1 (ii) The knowledge about animals is called _____.

Zoology

Q1 (iii) Bu Ali Sina is called _____ of the Muslim world.

Aristotle

Q1 (iv) _____ divided chemical compounds into four categories namely, minerals, botanicals, animals and derived compounds.

Al-Razi

Q1 (v) _____ Muslim scientist is considered as the pioneer in the field of chemistry.

Jabir Bin Hayyan

Q1 (vi) Kitab-ul-Manazir is the first book on _____.

Optics

Exercise 1 - Q2 True / False

Q2 (i) Bu Ali Sina was considered to be one of the founders of medicine. — True

Q2 (ii) Jabir Bin Hayyan was the first who discussed in detail the causes, symptoms, cure and prevention of smallpox and measles. — False

Q2 (iii) Jabir Bin Hayyan was an expert in physics. — False

Q2 (iv) Kitab-ul-Manazir is a publication of Al-Bairuni. — False

Q2 (v) The knowledge about animals is called botany. — False

Q2 (vi) In the life of animals and plants, many factors are common. — True

Exercise 1 - Q3 Multiple Choice

Q3 (i) Ibn-ul-Haitham is related to which branch of science?

(a) Sound

(b) Heat

✓ (c) Light

(d) Chemistry

Q3 (ii) The name of the famous book written by Al-Bairuni is

(a) Kitab-ul-Manazir

(b) Al-Havi

(c) Al-Shafa

✓ (d) Tahreer-ul-Amakin

Q3 (iii) The branch of science related to mechanics, heat, light and sound is

(a) Geology

(b) Astronomy

(c) Chemistry

✓ (d) Physics

Exercise 1 - Q4 to Q12 Short and Detailed Questions

Q4 What is meant by science?

The word science is derived from the Latin word Scientia, which means to study and analyse facts in their original form. Science is based on observation and hypothesis, and to establish a scientific law in the light of experiments is called the scientific method.

Q5 Write down the important branches of science. What do you know about each branch of science?

The important branches of science are physics, chemistry, biology, astronomy, mathematics, agriculture, medicine and geography. Physics deals with matter and energy; chemistry deals with the nature and composition of things; biology is the study of living things; astronomy studies the universe; and medicine deals with human anatomy, diagnosis and cure of diseases.

Q6 Write down the names of two Muslim scientists and give their important contribution to the service of science.

Jabir Bin Hayyan is said to be the founder of chemistry. He discovered methods to extract metals from ores, prepared nitric acid, sulphuric acid and hydrochloric acid, and was the first chemist who had his own laboratory. Ibn-ul-Haitham discovered the pinhole camera and wrote Kitab-ul-Manazir, a complete book on light, refraction, reflection, lenses and mirrors.

Q7 Write down the names of a few Pakistani scientists and discuss their contributions.

Dr. Abdus Salam was the only Pakistani scientist who won the Nobel Prize. In 1979 he won the Nobel Prize for his work in theoretical physics on the unification of the weak nuclear force and the electromagnetic force. Dr. Abdul Qadeer Khan directed the Kahuta Research Laboratories and, along with other Pakistani scientists, successfully tested nuclear devices at Chaghi on 28 May 1998, as a

result of which Pakistan became an atomic power.

Q8 What are the limitations of science?

Despite all achievements, science is still helpless in several matters. Genetic diseases are still incurable and the study of the genome is not complete; diseases like AIDS and hepatitis could not be controlled and cancer is still incurable. The food problem of mankind could not be solved completely, natural resources are exhausting, nuclear waste disposal is a problem, and natural disasters like earthquakes could not be predicted or controlled.

Q9 What is meant by technology? Give some examples of ancient technology.

Technology is the practical application of scientific knowledge to make tools, machines and articles of daily use. Examples of ancient technology are the wheel of the clay-man, the forge of the blacksmith, the spindle of the weaver, the rahat and plough of the farmer, and oarage boats.

Q10 Define biology. Discuss that it is a branch of science.

Biology is the study of living things. The word biology is derived from two Greek words, bios meaning life and logos meaning arguments. In biology we study the growth, working and reproduction of living things, including man, animals and plants. It is a branch of science because it studies living things systematically through observation and experiment, and it has two main branches: botany, the study of plants, and zoology, the study of animals.

Q11 The importance of science and knowledge has been discussed in the Holy Quran. Write two verses of the Holy Quran to explain.

The Holy Quran says: Read! In the name of thy Lord and Cherisher, Who created. Created man, out of a clot of congealed blood. Read!

And thy Lord is most Bountiful. He Who taught the use of the Pen. Taught man that which he knew not (Surah 96, verses 1-5). And Allah says: And of every thing We have created pairs: that ye may receive instruction (Surah 51, verse 49). These verses invite man to think, observe and seek knowledge, which is the basis of science.

Q12 What is physics? Write down the names of its important branches.

Physics is that branch of science which deals specially with matter and energy. It is also called the science of measurement because it deals mostly with measurement. Its main branches are mechanics, heat, light, sound and electricity.

Chapter 2: Our Life and Chemistry

Exercise 2 - Q1 Fill in the blanks

Q1 (i) _____ is the process by which plants produce glucose.

Photosynthesis

Q1 (ii) The percentage composition of methane is about _____ in natural gas.

90%

Q1 (iii) _____ is the only chemical compound found naturally as a liquid, as a solid, and as a gas.

Water

Q1 (iv) Nitrogen occurs in plants and animals in the form of _____.

Proteins

Q1 (v) Dilute solution of iodine in ethanol is called _____.

Tincture of iodine

Q1 (vi) Phosphorus is an important constituent of _____.

DNA, bones and teeth

Q1 (vii) Carbon is an important _____ for all organisms.

Element

Exercise 2 - Q2 Multiple Choice

Q2 (i) The form of carbon which is not crystalline is

✓ (a) Charcoal

(b) Graphite

(c) Bucky Ball

(d) Diamond

Q2 (ii) The process of converting atmospheric nitrogen to usable form is called

(a) Nitrogen Cycle

(b) Carbon Cycle

✓ (c) Nitrogen fixation

(d) Water cycle

Q2 (iii) Oxygen and nitrogen react to form

(a) Nitric acid

✓ (b) Nitrogen oxide

(c) Nitrogen peroxide

(d) Nitrate

Q2 (iv) The amount of carbon dioxide in the air is increased by

(a) Photosynthesis

(b) Respiration

✓ (c) Burning

(d) Evaporation

Q2 (v) The deficiency of iodine in human body causes

✓ (a) Goitre

(b) Cancer

(c) Tuberculosis

(d) Cholera

Q2 (vi) The contents of sodium in leaf tissues vary from

✓ (a) 0.01 - 10%

(b) 10 - 15%

(c) 16 - 12%

(d) 16 - 20%

Exercise 2 - Q3 to Q6 Short and Detailed Questions

Q3 (i) What is allotropy?

Allotropy is the property by which an element is found in more than one physical form in the same state. Carbon is the most common example, which exists in three allotropic forms: diamond, graphite and bucky balls.

Q3 (ii) Name those three elements which are abundantly found in human body.

The three elements abundantly found in the human body are carbon, hydrogen and oxygen. They combine to form organic compounds such as proteins, carbohydrates and lipids, which work as building material for the bodies of living organisms.

Q4 Why does water expand on freezing?

Water expands on freezing because its molecules are less tightly packed in ice than in liquid water. The density of ice at 0 C is 0.918 g/cm3 while that of water is about 1.00 g/cm3, so there is a considerable increase in volume when liquid water changes to solid ice. That is why ice floats on water and pipes can burst in winter.

Q5 Write notes on (1) Water as universal solvent (2) Properties of water.

Water is a universal solvent because it dissolves a variety of substances, so it is extensively used as a solvent in industrial chemical processes and many chemical reactions are carried out in it. Its properties are: it is a clear, colourless and odourless liquid; it has a low freezing point of 0 C and a high boiling point of 100 C; ice floats on water because its density is less than that of water; and its density is maximum at 4 C.

Q6 Write the importance of any two elements found in air.

Oxygen is the second most abundant gas in air and is essential for supporting life and the phenomena of burning and rusting. For burning, three things are required: fuel, heat and oxygen. Nitrogen is the major constituent of air; it is relatively inert and acts as an important diluent of air to slow down combustion and corrosion, and it occurs in plants and animals in the form of proteins.

Chapter 3: Biochemistry and Biotechnology

Exercise 3 - Q1 Fill in the blanks

Penicillins are made by a fungus called _____.

Penicillium

Human blood can be classified into A, B, AB and O groups on the basis of antigen and _____.

Antibody

Diabetes and haemophilia are due to change in _____ molecule.

Gene

The products of fat digestion are called _____.

Fatty acids and glycerol

Cephalosporins are obtained from fungus _____.

Cephalosporium

Exercise 3 - Q2 True / False

Metabolism is a combination of anabolic and catabolic reactions. — True

Fats are stored in the epithelial cells of human body. — False

Penicillin is a broad spectrum antibiotic. — False

Exercise 3 - Q3 Multiple Choice

The function of platelets is

(a) Blood clotting

(b) Engulfing the bacteria

(c) To produce antibodies

(d) Transfer of oxygen

Genetic information is transferred by

(a) Nucleus

(b) Chromosomes

✓ (c) Genes

(d) Gametes

Fats are formed by the combination of

(a) Glucose + glucose

(b) Water + carbon dioxide

✓ (c) Glycerol + fatty acids

(d) Amino acids + water

Penicillin was discovered by

(a) Robert Brown

✓ (b) Sir Alexander Fleming and Howard Florey

(c) Edward Jenner

(d) Robert Hooke

The antibiotic cephalosporin was discovered in

(a) 1848

✓ (b) 1948

(c) 1928

(d) 1998

Exercise 3 - Q4 Brief Answers

Write the names of three main types of cells in blood.

✓

The three main types of cells in blood are red blood cells, white blood cells and platelets. Red cells carry oxygen, white cells defend the body against germs, and platelets help the blood to clot.

Which tissues store fats in human body?

Fats are stored in the adipose tissue of the human body, under the skin and around the internal organs. This stored fat gives energy when food is short and also protects the organs and keeps the body warm.

What is transgenic organism?

A transgenic organism is an organism into which a gene from another organism has been transferred by genetic engineering. Such an organism shows the new character of the transferred gene, for example a crop plant that resists insects or disease.

What is a catalyst?

A catalyst changes the speed of a chemical reaction without undergoing any chemical change in its own structure. In living bodies, enzymes act as catalysts and speed up digestion and other life reactions without being used up.

Exercise 3 - Q5 to Q12 Detailed Questions

What is metabolism? Explain its different types.

Metabolism is the sum of all chemical reactions in a living body. It has two types: anabolism, in which simple substances join to build complex substances and energy is used, and catabolism, in which complex substances break down into simple ones and energy is released.

What is meant by digestion and assimilation of food?

Digestion is the breaking down of large, insoluble food molecules into small, soluble molecules that the body can absorb. Assimilation is the use of these absorbed food molecules by the body cells for growth, repair and energy.

What is an enzyme? What role is played by enzymes in our daily life?

An enzyme is a biological catalyst, made of protein, which speeds up a chemical reaction in the body without being changed itself. In daily life enzymes digest our food, help in making bread, cheese and curd, and are used in washing powders to remove stains.

What are the components of blood?

Blood has two main components: plasma, the liquid part, and blood cells. The cells are red blood cells which carry oxygen, white blood cells which fight germs, and platelets which clot the blood. Plasma carries food, hormones and waste materials.

How is DNA the hereditary material? Explain in detail.

DNA is the hereditary material because it carries the genetic information of an organism in the form of genes. When a cell divides, DNA copies itself and passes the same information to the new cells, so children inherit the characters of their parents through DNA.

What is genetic engineering? How can it help in improving agriculture and livestock?

Genetic engineering is a technique to transfer genes from one organism to another. In agriculture it produces crops that give more yield and resist insects, diseases and harsh weather, and in livestock it produces animals that give more milk and meat and resist disease.

What is an antibiotic? Write down its types.

An antibiotic is a chemical compound obtained from microorganisms and used to control infectious diseases caused by bacteria. Its types

include penicillins, made by the Penicillium fungus, and cephalosporins, obtained from the fungus Cephalosporium.

What is recycling? Give a detailed account of how waste and scarce materials can be made useable.

Recycling is producing new products from useless wastes or rubbish. Waste paper, glass, plastic and metal are collected, cleaned and remade into useful articles, which saves raw material and energy and keeps the environment clean. Scarce materials are thus used again instead of being thrown away.

Chapter 4: Human Health

Exercise 4 - Q1 Fill in the blanks

The most abundant naturally occurring carbohydrate of the world is _____.

Cellulose

Fats and oils are formed by chemical combination of fatty acids with _____.

Glycerol

Deficiency of vitamin _____ causes night blindness.

A

Goitre is caused by the deficiency of _____.

Iodine

Insulin and _____ are formed in pancreas.

Glucagon

Rabies is caused by _____ bite.

Dog

Exercise 4 - Q2 True / False

Glucose is the building block of proteins. — False

Vitamin A is a fat soluble vitamin. — True

Rickets is caused by vitamin C deficiency. — False

One gram of fat gives 4.1 kcal of energy. — False

Thyroxin hormone is secreted by parathyroid gland. — False

Exercise 4 - Q3 Multiple Choice

The compound which is required in very small amount is

(a) Carbohydrate

(b) Proteins

(c) Fats

✓ (d) Vitamins

The amount of energy gained from one gram of fats is

✓ (a) 9.3 kcal

(b) 18 kcal

(c) 27 kcal

(d) 36 kcal

The disease which is caused in children due to vitamin D deficiency is

(a) Scurvy

(b) T.B.

✓ (c) Rickets

(d) Anaemia

The hormone which controls involuntary action of body is

(a) Thyroxin

✓ (b) Epinephrine

(c) Adrenal

(d) Insulin

The disease caused by deficiency of iodine is

✓ (a) Goitre

(b) Night blindness

(c) Malaria

(d) Cough

Exercise 4 - Q4 Short Questions

What are the basic components of food?

The basic components of food are carbohydrates, lipids (fats and oils), proteins, vitamins, mineral salts and water. Carbohydrates and fats supply energy, proteins build and repair the body, and vitamins and minerals keep the body healthy and protect it from disease.

What is the role of vitamin B in body?

Vitamin B is a group of water soluble vitamins needed for healthy nerves, skin and blood and for releasing energy from food. Its deficiency causes diseases like beri-beri and anaemia, and it must be taken regularly in food because the body cannot store it.

What is the role of iron in the body?

Iron is needed to make haemoglobin in the red blood cells, which carries oxygen from the lungs to all parts of the body. Without enough iron the blood cannot carry sufficient oxygen, which causes anaemia, weakness and tiredness.

Which disease is caused by dog bite?

Rabies is caused by the bite of a rabid dog (and other rabid animals). The rabies virus attacks the nervous system, so the bite wound must be washed at once and anti-rabies vaccination started without delay.

What is the role of insulin?

Insulin is a hormone made in the pancreas which controls the amount of glucose in the blood. It helps the body cells take in glucose and store the extra glucose, and its deficiency causes diabetes, in which blood sugar rises to a harmful level.

Exercise 4 - Q5 to Q10 Detailed Questions

Write a note on important components of food.

Food has six important components. Carbohydrates are the primary source of energy for animals, while fats and oils give more energy and keep the body warm. Proteins, made of amino acids, build and repair tissues. Vitamins and mineral salts are needed in small amounts for health and protection from disease, and water carries food and waste through the body.

What do you know about protein?

Proteins are large compounds made of amino acids and are the building material of the body. They build muscles, skin, blood and enzymes, repair worn-out tissues and help growth. Meat, fish, eggs, milk, pulses and beans are rich sources of protein.

What are vitamins? Describe them in detail.

Vitamins are organic compounds needed in very small amounts to keep the body healthy and to protect it from deficiency diseases. Vitamins A, D, E and K are fat soluble, while B and C are water soluble. Vitamin A keeps eyesight healthy, D strengthens bones, C prevents scurvy, and B keeps nerves and blood healthy.

What is balanced diet? Describe the balanced diet for various people of our society.

A balanced diet is food that contains all the nutrients - carbohydrates, proteins, fats, vitamins, minerals and water - in the right proportions for health and growth. Growing children need more protein and calcium, hard-working labourers need more carbohydrates for energy, and old people need light, easily digested food with enough vitamins and minerals.

Why is exercise important for health?

Exercise is necessary to maintain human health because it

strengthens muscles, heart and lungs and keeps the body active and fit. It improves blood circulation and digestion, controls body weight and removes tiredness, and it also freshens the mind and helps sound sleep.

Discuss the role of various endocrine glands in our body.

Endocrine glands bring about coordination in the body through hormones carried by the blood. The thyroid gland controls growth and metabolism through thyroxin, the pancreas controls blood sugar through insulin and glucagon, and the adrenal glands prepare the body for emergency action through epinephrine. The pituitary gland controls the working of other glands.

Chapter 5: Diseases, Cause and Prevention

Exercise 5 - Q1 Fill in the blanks

Bacteria can be seen by a _____.

Microscope

E.P.I. stands for _____.

Expanded Programme on Immunization

The AIDS virus is called _____.

HIV

The measles vaccination is given to a child at _____.

9th month

Hepatitis virus spreads by faeco-_____ route through food and water.

Oral

B.C.G injection protects from _____.

Tuberculosis

Exercise 5 - Q2 True / False

Polio virus affects the nervous system. — True

Antibiotic drugs are used against the viral diseases. — False

Tuberculosis is a non treatable disease. — False

AIDS is not an infectious disease. — False

Cigarette smoker remains protected from lung and heart diseases. — False

Exercise 5 - Q3 Multiple Choice

The age of child in which the child is injected with measles is

(a) At birth

(b) First month

(c) 3rd month

✓ (d) 9th month

The drinks which can be used more in hepatitis are

(a) Water

(b) Sugarcane juice

(c) Juices

✓ (d) All

The age of the child in which he is given first injection of BCG is

(a) First month

✓ (b) At Birth

(c) Third month

(d) Ninth month

BCG saves the child from

(a) Measles

(b) Whooping cough

✓ (c) T.B.

(d) Hepatitis

DPT is not effective in

(a) Diphtheria

✓ (b) Polio

(c) Whooping cough

(d) Tetanus

The chemical present in smoke of cigarette which makes the person

addict is

(a) Tar

(b) Nicotine

(c) Carbon monoxide

(d) Nitrogen dioxide

Exercise 5 - Q4 Short Questions

In which age is a child injected against measles and why?

A child is vaccinated against measles at the age of 9 months. By this age the protection received from the mother has weakened, so the vaccine is needed to build the child's own immunity against measles.

What is the name of AIDS virus?

The AIDS virus is called HIV, which stands for Human Immuno Deficiency Virus. It damages the immune system of the body, so the body can no longer defend itself against diseases.

In which diseases does DPT injection give immunity to the body?

DPT injection gives immunity against three diseases: diphtheria, whooping cough (pertussis) and tetanus. DPT stands for the vaccines against these three diseases given together in childhood.

How does malaria spread?

Malaria spreads through the bite of the female Anopheles mosquito carrying the malaria parasite. When the infected mosquito bites a healthy person, the parasite enters the blood, attacks the red blood cells and causes fever with shivering.

Write the names of different agents causing diseases.

Diseases are caused by different agents: bacteria, viruses, fungi, protozoa and worms. Mosquitoes, Ascaris and thread worms can also

cause many diseases, and germs spread by air, water, touch, faeces and through animals.

What is sterilization?

Sterilization is the process of killing all germs from instruments, dressings and other articles by heat, boiling or chemicals. It prevents germs from entering wounds during treatment and stops the spread of infection in hospitals.

Exercise 5 - Q5 to Q11 Detailed Questions

How does AIDS spread? Write its preventive measures.

AIDS spreads through infected blood, by using infected needles and syringes, from an infected mother to her child, and through unprotected contact with an infected person. It does not spread by eating, shaking hands or living together. Prevention requires tested blood for transfusion, sterilized syringes, and public awareness about its causes and safety measures.

Write different methods to prevent from malaria.

Malaria is prevented by avoiding mosquito bites and by destroying mosquitoes. Use mosquito nets and repellents, cover water containers, drain standing water, spray insecticides in houses and streets, and take complete anti-malarial treatment at once if fever appears.

What are the effects of smoke and smoking?

Cigarette smoke contains many dangerous gases, including nicotine, tar and carbon monoxide, which cause diseases of the lungs and heart. Smoking makes a person addicted through nicotine, damages the lungs, raises the risk of cancer and heart disease, and harms the

health of people nearby who breathe the smoke.

Briefly describe some of the mental diseases.

Mental diseases affect thinking, feeling and behaviour, and they must be treated like other diseases. They include severe anxiety and depression, in which a person feels constant fear or sadness, and more serious disorders in which a person loses touch with reality. Proper treatment, care and sympathy help such patients recover.

What do you know about Dengue Fever?

Dengue fever is a viral disease spread by the bite of the Aedes mosquito, which breeds in clean standing water. Its symptoms are sudden high fever, severe headache, pain in the joints and muscles, and skin rash. There is no specific medicine for it, so the patient needs rest, fluids and medical care.

Describe treatment and prevention of Dengue Fever.

Dengue fever is treated by rest, plenty of fluids and medicines for fever and pain given on a doctor's advice, with hospital care in serious cases. It is prevented by destroying Aedes mosquitoes: cover water containers, remove standing water, use nets and repellents, and keep the surroundings clean.

What is Dengue Haemorrhagic Fever (DHF)? Write in detail.

Dengue Haemorrhagic Fever is the severe form of dengue in which blood vessels leak and bleeding occurs from the nose, gums and skin, with a dangerous fall in platelets. The patient feels severe pain in the abdomen, vomits repeatedly and becomes weak, so urgent hospital treatment with fluids and blood care is needed to save life.

Chapter 6: Environment and Natural Resources

Exercise 6 - Q1 Fill in the blanks

Q1 (i) Atmosphere is an _____ of gases that covers the earth from all sides.

Envelope

Q1 (ii) Ozone prevents _____ from reaching the earth.

Ultraviolet rays

Q1 (iii) Temperature of the thermosphere may be up to _____.

2000 C

Q1 (iv) Radiation of _____ wavelength cannot go out of greenhouse.

Long

Q1 (v) Substances that cause pollution of the environment are called _____.

Pollutants

Q1 (vi) Coal, oil and gas are called _____.

Fossil fuels

Q1 (vii) Fossil fuels and minerals are _____ resources.

Non-renewable

Q1 (viii) One reason for the extinction of many species is _____ destruction.

Habitat destruction

Q1 (ix) Areas reserved for wildlife are called _____.

Wildlife reserves

Q1 (x) Moving out from an area and settling in another area is called

_____.

Migration

Exercise 6 - Q2 Multiple Choice

Q2 (i) Thickness of atmosphere is

✓ (a) 200 km

(b) 1000 km

(c) 1200 km

(d) 1600 km

Q2 (ii) The proportion of carbon dioxide in air is

(a) 40 percent

(b) 0.4 percent

✓ (c) 0.04 percent

(d) 0.004 percent

Q2 (iii) Ozone forms a protective layer in

(a) Troposphere

✓ (b) Stratosphere

(c) Mesosphere

(d) Thermosphere

Q2 (iv) Major cause of depletion of ozone layer is

(a) Oxygen

(b) Hydrogen

✓ (c) Chlorofluorocarbons

(d) Hydrocarbon

Q2 (v) About _____ percent population of Pakistan is dependent upon agriculture

(a) 90%

(b) 80%

✓ (c) 60%

(d) 50%

Q2 (vi) The total number of persons living in an area is called

(a) Species

✓ (b) Population

(c) Community

(d) Habitat

Q2 (vii) Pakistan's population in 1998 was

✓ (a) 130.5 Million

(b) 13.5 Million

(c) 135 Million

(d) 0.135 Million

Q2 (viii) At present population growth rate is 2.6 percent, in how many years will the population of Pakistan double?

(a) 47 years

(b) 37 years

✓ (c) 27 years

(d) 17 years

Exercise 6 - Q3 Short Questions

Q3 (i) Define (a) Pollution (b) Pollutants (c) Recycling (d) Endangered species.

Pollution is the addition of harmful substances to the environment. Pollutants are the substances that cause pollution of the environment. Recycling is the production of new useful articles from used materials. Endangered species are those animals and plants

which are in danger of extinction, like the snow leopard, Indus river blind dolphin and Balochistan bear in Pakistan.

Q3 (ii) Write names of the four layers of atmosphere.

The four layers of the atmosphere are the troposphere, the stratosphere, the mesosphere and the thermosphere. The troposphere extends to about 18 km, the stratosphere to 50 km, the mesosphere to about 85 km, and the thermosphere is the outermost and hottest layer.

Q3 (iii) Write any two effects of greenhouse effect on environment.

Two effects of the greenhouse effect on the environment are: the earth's climate and weather may change, and melting of ice caps on the poles and mountains and excessive rains may raise the sea level, so many coastal areas may go under sea.

Q3 (iv) Write two methods of conservation of natural resources.

Two methods of conservation of natural resources are recycling of used materials and reuse of articles of common use, and substitution, that is, use of alternatives. We should also make minimum and only necessary use of resources and not waste them.

Q3 (v) Write two benefits of wildlife.

Two benefits of wildlife are that wildlife maintains the balance of nature and the beauty of the environment, and it is a source of food, medicines and other useful materials for man. National parks and wildlife reserves also provide places for education and research.

Exercise 6 - Q4 to Q12 Detailed Questions

Q4 Explain composition and different layers of atmosphere.

The atmosphere is an envelope of gases that covers the earth from

all sides and is about 200 km thick. It is made up of about 78 percent nitrogen and 21 percent oxygen, and the remaining one percent consists of water vapour and trace gases like carbon dioxide, hydrogen, argon, helium and ozone. Its four layers are the troposphere, where most weather occurs; the stratosphere, whose upper part contains the ozone layer; the mesosphere, the coldest layer; and the thermosphere, the outermost and hottest layer, where temperature may be as high as 2000 C.

Q5 Write a note on depletion of ozone layer.

Ozone is a gas present in the upper stratosphere. It forms a protective covering around the earth, which prevents ultraviolet radiations from reaching the earth. Certain chemicals released from refrigerators, air conditioners, spray cans and factories making packing foams are called chlorofluorocarbons (CFCs). CFCs react with ozone, causing its depletion and thinning of the ozone layer. Consequently a greater number of ultraviolet radiations reach the earth, and these radiations may cause cancer and eye diseases.

Q6 What is meant by greenhouse effect? Describe its causes and effects on environment.

The greenhouse effect is the phenomenon in which solar radiations of short wavelength enter the atmosphere but heat waves of long wavelength cannot go out, so the temperature inside rises. Its causes are the increase in greenhouse gases like carbon dioxide, chlorofluorocarbons and methane in the air due to industrialization and burning of fossil fuels. Its effects are change in the earth's climate and weather, melting of ice caps and rising sea level, and flooding of coastal areas.

Q7 How do human activities affect environment?

Human activities affect the environment in many ways. Burning of fossil fuels increases carbon dioxide and produces smoke and gases that pollute the air. Industrial wastes, trash and sewage discharged into water bodies cause water pollution, and municipal trash and chemicals cause land pollution. Cutting of forests destroys natural habitats, and rising population increases the demand for food, housing and energy, which puts resources under pressure and lowers the quality of life.

Q8 Write down causes, effects and measures of controlling aquatic pollution.

Aquatic pollution is caused when industrial wastes, trash and sewage from cities are discharged into rivers, streams, ponds, lakes and oceans, and when oil spills from tankers and chemical fertilizers and pesticides washed off from crops enter the water. Its effects are the decrease of oxygen in water, death of fish and aquatic plants, and diseases like cholera and typhoid in man. It is controlled by treating domestic, industrial and hospital wastes before throwing them into water, by not throwing waste articles into water bodies, and by enforcing minimum environmental standards on industries.

Q9 Explain use and environmental effects of fossil fuels.

Fossil fuels are coal, oil and gas, formed from the remains of plants and animals of the remote past buried under the earth. They are used for transport, power generation, agriculture and industry, and natural gas is used in power plants, cement and fertilizer production, and in homes for heating and cooking. Their environmental effects are that burning of fossil fuels produces smoke and gases which pollute the air and increase carbon dioxide, and large areas of fertile land, forests and natural habitats are destroyed during exploration

and drilling of coal and oil.

Q10 Write a note on conservation of resources.

Conservation of resources means the careful use of natural resources so that they last for future generations. Fossil fuels and minerals are non-renewable resources because they cannot be reproduced or take millions of years to reform, and they may end up soon if their excessive use continues. They are conserved by recycling used materials, by substitution with alternatives, by reuse of articles of common use, and by making minimum and only necessary use of resources.

Q11 Write brief notes on (a) Mechanized farming and modern trends in agriculture (b) Dairy, poultry and fish farming (c) Wildlife conservation and National parks (d) Importance of wildlife.

Mechanized farming means cultivation with tractors, tube wells, harvesters and threshers instead of only muscle power, which increases crop yield and production for export and earning money. Dairy, poultry and fish farming provide milk, meat and eggs and are an important part of agriculture in Pakistan. Wildlife conservation means saving wildlife from extinction by re-establishing habitats and marking certain areas as wildlife reserves and national parks, where human interference is strictly prohibited except for education and research. Wildlife is important because it maintains the balance of nature and provides food, medicines and beauty to the environment.

Q12 Explain environmental problems arising due to increase in population.

Increase in population causes environmental problems of physical, economic and social nature. Basic necessities of life like clean air, water, housing and food are not properly fulfilled, and facilities of

health and education are not available for all. Resources come under pressure and development stops, poverty and poor quality of life increase, forests are destroyed, cities expand, and problems like crime, hunger and sense of deprivation cast very negative effects on society. Migration from rural to urban areas leads to a huge increase in city population and forces many people to live in slums.

Chapter 7: Energy

Exercise 7 - Q1 Multiple Choice

Q1 (i) The unit of energy is

(a) newton

(b) metre

✓ (c) joule

(d) second

Q1 (ii) The energy due to motion is called

(a) potential energy

✓ (b) kinetic energy

(c) nuclear energy

(d) chemical energy

Q1 (iii) The method of production of electricity that does not produce pollution is

✓ (a) hydroelectric power

(b) thermal power

(c) nuclear power

(d) burning of biogas

Q1 (iv) By burning fossil fuels, we get

(a) solar power

(b) tidal power

(c) nuclear power

✓ (d) thermal power

Q1 (v) We can save energy

(a) by increasing personal vehicles

(b) by making vehicles of heavier bodies

(c) by not walking

✓ (d) by avoiding unnecessary use of energy

Exercise 7 - Q2 Fill in the blanks

Q2 (i) Work is the product of force and distance travelled in the direction of _____.

Force

Q2 (ii) We can see things with the help of _____.

Light

Q2 (iii) Photocells convert light into _____.

Electricity

Q2 (iv) The energy of the sea tides is called _____ energy.

Tidal

Q2 (v) The safe disposal of _____ is very necessary to protect from radiation.

Nuclear waste

Exercise 7 - Q3 True / False

Q3 (i) The energy possessed by a body due to position is called kinetic energy. — False

Q3 (ii) Electrical energy is the energy of flowing charges. — True

Q3 (iii) Chemical potential energy is stored in fossil fuels. — True

Q3 (iv) Cooling towers of nuclear reactors absorb heat from the atmosphere. — False

Q3 (v) When pollution less than a certain limit is added to the environment, it is known as degradation of environment. — False

Exercise 7 - Q4 Short Answers

Q4 (i) What is the law of conservation of energy?

The law of conservation of energy states that energy can neither be created nor destroyed, but it can be changed from one form to another. For example, when a body is taken to a height, gravitational potential energy is stored in it, and when it is dropped, this energy converts into kinetic energy.

Q4 (ii) Define the unit of electrical energy.

The unit of electrical energy is the kilowatt-hour. One kilowatt-hour is the energy consumed when an appliance of one kilowatt power is used for one hour, and it is the unit in which electricity bills are charged.

Q4 (iii) Define environment.

Environment is the living place of people, including the physical and social factors around them. It includes air, water, land, plants, animals and the social conditions in which people live.

Q4 (iv) What is meant by degradation of environment?

Degradation of environment means the addition of pollution to the environment more than a certain limit, so that the quality of air, water and land falls and the life of plants, animals and man is harmed.

Q4 (v) Write down the names of three conventional methods and five non-conventional methods of producing electricity.

Three conventional methods of producing electricity are hydroelectric

power, thermal power and nuclear power. Five non-conventional methods are solar energy, wind energy, tidal energy, geothermal energy and biogas energy.

Q4 (vi) Give any three suggestions for the conservation of energy.

Three suggestions for the conservation of energy are: avoid unnecessary use of energy, use energy-saving appliances and bulbs, and use public transport and walk for short distances instead of using personal vehicles.

Q4 (vii) Point out two suggestions for the safe disposal of nuclear waste.

Two suggestions for the safe disposal of nuclear waste are: bury the waste deep underground in sealed containers in safe places away from people and water, and keep it under strict watch so that its radiations cannot reach living things.

Exercise 7 - Q5 to Q12 Detailed Questions

Q5 Define energy. What is the difference between kinetic and potential energy? Explain with the help of examples.

Energy is the ability of a body to do work. Kinetic energy is the energy possessed by a body due to its motion, for example a running cricket ball or flowing water. Potential energy is the energy possessed by a body due to its position, for example a stone lying at a height or water stored in a dam. The moving ball can do work because of its motion, while the stone at a height can do work when it falls.

Q6 Describe different forms of energy.

The different forms of energy are kinetic energy, potential energy, elastic potential energy, chemical energy, heat energy, light energy,

electrical energy and nuclear energy. Heat energy is due to the movement of molecules, light energy helps us see things, electrical energy is the energy of moving charges, and nuclear energy is obtained by breaking the nuclei of heavy atoms or by fusion of small nuclei.

Q7 What is meant by interconversion of energy? State law of conservation of energy.

Interconversion of energy means the change of energy from one form to another in different conditions. For example, the chemical energy in cells or batteries converts into electrical energy, and the electrical energy in a bulb converts into heat and light. The law of conservation of energy states that energy can neither be created nor destroyed, it only changes its form.

Q8 What are the conventional methods of producing electricity? Write a detailed note on any one of them.

The conventional methods of producing electricity are hydroelectric power, thermal power and nuclear power. In hydroelectric power, water stored in a dam is allowed to fall on turbines, which run generators and produce electricity. This method does not produce pollution, and the water can be used again for irrigation after producing electricity.

Q9 Describe any three non-conventional methods of producing electricity.

Three non-conventional methods of producing electricity are solar energy, in which photocells convert light into electricity; wind energy, in which windmills run generators; and tidal energy, in which the energy of sea tides runs turbines. Other methods are geothermal energy from the heat inside the earth and biogas energy from

biomass.

Q10 In what unit is electrical energy measured? Describe the principle and working of electricity meter.

Electrical energy is measured in kilowatt-hours, which is the unit shown on the electricity meter. The meter works on the principle that when current passes through its coils, a magnetic field is produced which rotates an aluminium disc. The speed of the disc depends on the power being used, and the number of revolutions of the disc is counted to show the total energy consumed in kilowatt-hours.

Q11 What is thermal pollution? How is it produced? How does it affect the environment?

Thermal pollution is the addition of heat and harmful gases to the atmosphere. It is produced when hot water from power stations and factories is thrown into rivers and when fossil fuels are burnt, which raises the temperature of air and water. It affects the environment by disturbing aquatic life, reducing oxygen in water and increasing the temperature of the surroundings.

Q12 Write notes on (i) Nuclear fuel hazards (ii) Conservation of energy.

Nuclear fuel hazards are the dangers of radiations from nuclear fuel and nuclear waste, which can cause cancer and other diseases and harm living things for a long time. Nuclear waste must be disposed of safely in sealed containers deep underground. Conservation of energy means avoiding unnecessary use of energy, using energy-saving devices, and using public transport, so that energy resources last longer and pollution is reduced.

Chapter 8: Current Electricity

Exercise 8 - Q1 True / False

Q1 (i) Amount of charge passing through any cross-section in one second is called electric current. — True

Q1 (ii) The unit of electric current is volt. — False

Q1 (iii) Current flows from higher potential to lower potential. — True

Q1 (iv) Switches should always be placed in the way of live wire. — True

Q1 (v) D.C. voltage can be decreased or increased with the help of a transformer. — False

Exercise 8 - Q2 Fill in the blanks

Q2 (i) The current flowing through a conductor can be considered analogous to water flowing through a _____.

Pipe

Q2 (ii) Potential difference is also called _____.

Voltage

Q2 (iii) The _____ to the flow of charges is called resistance.

Opposition

Q2 (iv) The charge on a capacitor is proportional to _____.

Potential difference

Q2 (v) Ammeter is always connected in _____ with the circuit.

Series

Exercise 8 - Q3 Multiple Choice

Q3 (i) The instrument that measures current is called

(a) voltmeter

(b) circuit breaker

✓ (c) ammeter

(d) switch

Q3 (ii) The potential of the neutral wire is

✓ (a) zero

(b) +220 Volts

(c) 220 Volts

(d) changing

Q3 (iii) The SI unit of resistance is

(a) ampere

(b) volt

(c) hertz

✓ (d) ohm

Q3 (iv) The constant in Ohm's law is

(a) current

✓ (b) resistance

(c) potential difference

(d) charge

Q3 (v) The device used for turning a circuit ON or OFF is

✓ (a) switch

(b) fuse

(c) circuit breaker

(d) earthwire

Exercise 8 - Q4 Short Answers

Q4 (i) What is an electric current?

Electric current is the amount of charge passing through any cross-section of a conductor in one second. Its unit is the ampere, and it is measured by an ammeter connected in series with the circuit.

Q4 (ii) Define potential difference.

Potential difference is the energy consumed by one coulomb of charge in moving from one point to another. It is also called voltage, and its unit is the volt. It is measured by a voltmeter connected in parallel with the circuit.

Q4 (iii) Which quantities does Ohm's law relate?

Ohm's law relates three quantities: potential difference, current and resistance. It states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided its temperature remains constant, and resistance is the constant in this relation.

Q4 (iv) Define resistance.

Resistance is the opposition to the flow of charges through a conductor. Its SI unit is the ohm. Conductors of large resistance are called resistors.

Q4 (v) What are resistors?

Resistors are conductors of large resistance, used in circuits to control the current. They are used in radios, televisions and other electrical appliances to reduce current to the required value.

Q4 (vi) What is meant by charging of a capacitor?

Charging of a capacitor means storing charge on its plates. When a

capacitor is connected to a battery, charge flows to its plates until the potential difference between the plates becomes equal to the battery voltage, and the charge stored is proportional to the potential difference.

Q4 (vii) Where are the gang capacitors used?

Gang capacitors are used in radios for tuning. By changing the capacitance of the gang capacitor, the radio is tuned to the required station, because the tuning circuit then receives only the signals of that frequency.

Q4 (viii) What does a step-up transformer do?

A step-up transformer increases the alternating voltage. It has more turns in its secondary coil than in its primary coil, so the output voltage is higher than the input voltage.

Q4 (ix) What does a step-down transformer do?

A step-down transformer decreases the alternating voltage. It has fewer turns in its secondary coil than in its primary coil, so the output voltage is lower than the input voltage.

Q4 (x) Where is a shunt resistance connected to convert a galvanometer into ammeter?

A shunt resistance is connected in parallel with a galvanometer to convert it into an ammeter. Most of the current passes through the low shunt resistance, while only a small safe current passes through the galvanometer, so the galvanometer can measure large currents.

Exercise 8 - Q5 to Q11 Detailed Questions

Q5 Define electric current. What is conventional current? Explain.

Electric current is the flow of charge through a conductor, measured

as the amount of charge passing through any cross-section in one second. Conventional current is the flow of positive charge from higher potential to lower potential. In a metal wire electrons actually flow from lower to higher potential, but by convention the current is taken as flowing from the positive terminal to the negative terminal.

Q6 Explain potential difference with example. Define its unit.

Potential difference between two points is the energy consumed by one coulomb of charge in moving from one point to the other. For example, if one joule of energy is consumed by one coulomb of charge in moving between two points, the potential difference is one volt. Its unit is the volt, and it is measured by a voltmeter.

Q7 What is resistance? Define the unit of resistance.

Resistance is the opposition offered by a conductor to the flow of charges. The unit of resistance is the ohm. One ohm is the resistance of a conductor when a potential difference of one volt across its ends causes a current of one ampere to flow through it.

Q8 What is a capacitor? Give some uses of capacitors.

A capacitor is a device that stores charge. It consists of two plates separated by an insulator, and the charge stored on it is proportional to the potential difference between its plates. Capacitors are used in radios for tuning, in fans and motors for starting, in flash cameras, and in power supply circuits for smoothing.

Q9 What is the difference between direct current and alternating current? Explain.

Direct current is current that flows in one direction only, for example current from a cell or battery. Alternating current is current that changes its direction again and again, for example the current supplied to our homes. Direct current voltage cannot be changed by

a transformer, while alternating voltage can be increased or decreased by a transformer.

Q10 How does structure of a voltmeter differ from that of an ammeter? Describe the use of both of them.

A voltmeter is connected in parallel with a circuit and has a very high resistance, so it takes almost no current and measures potential difference. An ammeter is connected in series with a circuit and has a very low resistance, so it does not disturb the current and measures the current flowing through the circuit.

Q11 What is multimeter? What is it used for? Describe the difference in analogue and digital meter.

A multimeter is an instrument for measuring current, voltage and resistance. It is used to test circuits and appliances and to find faults in them. An analogue meter shows a continuous reading with the help of a pointer on a scale, while a digital meter displays the reading in digits on a screen.

Chapter 9: Basic Electronics

Exercise 9 - Q1 Multiple Choice

Q1 (i) In p-type semi-conductor, most of the current is due to

(a) free electrons

✓ (b) holes

(c) positive ions

(d) heat

Q1 (ii) Diodes are used to

✓ (a) convert A.C. into D.C.

(b) convert D.C. into A.C.

(c) store charge

(d) change voltage

Q1 (iii) Electric signal is converted into digital signal by

(a) keyboard

(b) monitor

(c) scanner

✓ (d) modem

Q1 (iv) In binary system, 37 is written as

(a) 101101

✓ (b) 100101

(c) 110011

(d) 101011

Q1 (v) Analogue signal is recorded on

✓ (a) magnetic tape

(b) floppy disk

(c) hard disk

(d) C.D.

Exercise 9 - Q2 Fill in the blanks

Q2 (i) Electronics is the knowledge of behaviour and control of _____.

Electric current

Q2 (ii) A _____ beam scans C.D. for replay.

Laser

Q2 (iii) Programme is a list of _____.

Instructions

Q2 (iv) The orbit of a hovering satellite is known as _____ orbit.

Geostationary

Q2 (v) In the picture tube, an electron gun throws a beam of _____ on the screen.

Electrons

Exercise 9 - Q3 True / False

Q3 (i) The quality of sound from digital recording of C.D. is much better as compared to that of cassette tape. — True

Q3 (ii) Analogue signal consists of distinct on/off electrical pulses. — False

Q3 (iii) In cable T.V. electrical signals are converted into radio waves. — False

Q3 (iv) Fax machine is used to send and receive documents and pictures from one place to another. — True

Q3 (v) At least four hovering satellites are required to send transmissions all over the world. — False

Exercise 9 - Q4 to Q10 Detailed Questions

Q4 What are n-type and p-type semi-conductors? Where are they used for?

An n-type semi-conductor is a semi-conductor in which current is mostly due to free electrons, obtained by doping silicon with a pentavalent impurity. A p-type semi-conductor is one in which current is mostly due to holes, obtained by doping silicon with a trivalent impurity. They are used to make diodes, transistors and other electronic devices like radios, televisions and computers.

Q5 How is a diode forward biased and reverse biased? Describe different kinds of diodes and write few uses of them.

A diode is forward biased when its p-side is connected to the positive terminal and n-side to the negative terminal of a battery, so current flows easily. It is reverse biased when the connections are reversed, so almost no current flows. Different kinds of diodes are the P-N junction diode used as a rectifier to convert A.C. into D.C., the light emitting diode used in displays and indicators, and the photo diode used in light-measuring devices.

Q6 What are radiowaves? How do radio transmissions reach us?

Radio waves are a type of electromagnetic waves, also called carrier waves. Radio transmissions reach us when sound is converted into electrical signals, which are carried by radio waves from a transmitter through the air to a receiver. The receiver separates the sound signals from the carrier waves and changes them back into

sound through a speaker.

Q7 How does television work? Explain briefly the satellite T.V.

Television works by converting pictures and sound into electrical signals at the transmitting station, sending them through space or cable, and changing them back into pictures and sound at the receiver. In satellite T.V., signals are transmitted to a geostationary satellite through microwaves by a ground station. The satellite retransmits them to the required ground station after amplifying them, from where they are transmitted onwards to homes.

Q8 Write an explanatory note on television.

Television is an electronic system for sending pictures and sound from one place to another. The camera changes pictures into electrical signals and the microphone changes sound into electrical signals. These signals are transmitted by radio waves or cable to receivers, where the picture tube changes the video signals back into pictures on the screen and the speaker changes the audio signals back into sound.

Q9 What are the main parts of a computer and what do they work?

The main parts of a computer are the input devices, the central processing unit, the output devices and the information storage devices. Input devices like the keyboard feed data into the computer; the central processing unit, including the microprocessor, processes the data; output devices like the monitor and printer display or print the results; and storage devices like the hard disk, floppy disk and C.D. store data for later use.

Q10 Write a note on communication system.

A communication system is an electronic method to transmit information from one place to another. Its main sources are radio,

television, telephone, fax machine, computer and Internet. Information is transmitted after converting it into electrical signals, which are sent through wires, radio signals through air, or light signals through optical fibres. Telecommunication is the method used for instant communication of information to far off places.

Chapter 10: Science and Technology

Exercise 10 - Q1 True / False

Q1 (i) There is no difference in laser light and ordinary light. — False

Q1 (ii) Hearing aid is called radar. — False

Q1 (iii) X-rays can pass through flesh. — True

Q1 (iv) Rays emitted from radioactive sources are harmless. — False

Q1 (v) Ultrasound are the sound waves of frequency greater than 20 kHz. — True

Exercise 10 - Q2 Multiple Choice

Q2 (i) Emission of radiations from nucleus is said to be

(a) chemical reaction

(b) atomic reaction

✓ (c) radioactivity

(d) nuclear fission

Q2 (ii) Frequency of ultrasound is

(a) less than 20 Hz

(b) 20 Hz

(c) 20 kHz

✓ (d) more than 20 kHz

Q2 (iii) Radioactivity occurs naturally from all the elements with atomic number greater than

(a) 62

(b) 70

(c) 80

✓ (d) 82

Q2 (iv) The principle of light on which the fibre optics work is

(a) reflection

(b) refraction

✓ (c) total internal reflection

(d) dispersion

Exercise 10 - Q3 Fill in the blanks

Q3 (i) Alpha particles are deflected towards _____ plate.

Negative

Q3 (ii) The _____ rays are not affected by any field.

Gamma

Q3 (iii) Optical fibres are fine _____ of glass.

Fibres

Q3 (iv) The _____ of all waves of laser is the same.

Wavelength

Q3 (v) E.C.G. evaluates the _____ activity of heart.

Electrical

Exercise 10 - Q4 Brief Answers

Q4 (i) Which objects are used for laser?

Crystals like ruby and glass, semi-conductors and some gases are used to make lasers. The first laser ever used was a ruby crystal, which was excited by a powerful flash of light. Two plane mirrors are

used to amplify the light again and again.

Q4 (ii) From where do the satellites get electrical power?

Satellites get electrical power from solar panels fitted on them. The solar cells of the panels convert sunlight directly into electricity, which runs the instruments of the satellite and charges its batteries for use when the satellite is in the shadow of the earth.

Q4 (iii) What are called the elements, which emit radiation?

The elements which emit radiations are called radioactive elements. They occur naturally among elements with atomic number greater than 82, and their nuclei emit alpha, beta and gamma radiations.

Q4 (iv) Differentiate between E.C.G. and E.E.G.

E.C.G. evaluates the electrical activity of the heart, and is used to find heart diseases. E.E.G. evaluates the electrical activity of the brain, and is used to find brain diseases. Both record the electrical waves of the organ on a graph, but one is for the heart and the other for the brain.

Q4 (v) How is steel made harder?

Steel is made harder by adding carbon to iron and by heat treatment. When steel is heated and then suddenly cooled, it becomes hard, and this process is used to make tools, blades and other hard steel articles.

Exercise 10 - Q5 to Q18 Detailed Questions

Q5 What is laser? Describe few important uses of it.

Laser is an abbreviation of light amplification by stimulated emission of radiation. It is an intense beam of light in which all the waves have the same wavelength and all are in phase, and it travels in one

direction without spreading. Its important uses are: in surgery as a light scalpel for cutting and coagulation, in ophthalmology for cataract and glaucoma operations, in dermatology for skin diseases, in dentistry, in treating cancer and crushing gallstones and kidney stones, in drilling holes in hard materials like steel and diamond, in making holograms, and in laser-guided missiles and communication.

Q6 Define optical fibre. Describe its construction, principle and working.

An optical fibre is a fine fibre of glass used to carry light signals. In construction it has a central core of glass surrounded by a cladding of glass of lower refractive index, covered by a protective coating. Its principle is total internal reflection: when light enters one end of the fibre at a suitable angle, it is reflected again and again inside the core and comes out of the other end without loss. In working, light signals carrying information are sent into the fibre and travel through it by repeated total internal reflection to the other end.

Q7 What do you mean by radar system? How does it work? Write some of its uses.

Radar system means radio detection and ranging. It works by sending radio waves towards an object and receiving the waves reflected back from it. From the time taken by the waves to return, the distance, direction and speed of the object are found. Its uses are in airports for guiding aeroplanes, in ships for finding other ships and icebergs, in the army for detecting enemy aeroplanes and missiles, and in weather forecasting.

Q8 Discuss in detail the satellite and its types and highlight its uses.

A satellite is an object placed in orbit around the earth to receive signals from a ground station and retransmit them after amplifying.

Its types are communication satellites, which facilitate television, telephone and radio communication; weather satellites, which help in weather prediction; and scientific satellites, which study the universe, galaxies, stars and planets. Their uses are in communication, weather forecasting, exploration of natural resources, television transmission and scientific research.

Q9 What is radioactivity? How many are the types of radiations? Describe their characteristics.

Radioactivity is the emission of radiations from the nucleus of an atom. There are three types of radiations. Alpha radiations are positively charged particles, deflected towards the negative plate, and they have low penetrating power. Beta radiations are negatively charged particles, deflected towards the positive plate, and they have greater penetrating power than alpha rays. Gamma radiations have no charge, are not affected by any field, and have the greatest penetrating power.

Q10 What are radioisotopes? Write down some of their uses.

Radioisotopes are the radioactive isotopes of elements, whose nuclei emit radiations. Their uses are in medicine for diagnosis and treatment of diseases like cancer, in agriculture for producing better varieties of crops and preserving food, in industry for checking the thickness of materials and finding faults, and in research for tracing chemical reactions.

Q11 How are X-rays obtained? Write their properties and uses.

X-rays are obtained when fast-moving electrons strike a metal target in an X-ray tube, which produces high-energy radiations. Their properties are that they can pass through flesh but are stopped by bones and dense materials, they affect photographic film, and they

travel in straight lines. Their uses are in medicine for seeing broken bones and internal organs, in industry for finding faults inside materials, and in security checking of luggage.

Q12 What is the difference between X-rays and C.T. scan? Which method is better for treatment?

X-rays give a flat, two-dimensional picture of the inside of the body on a film, while a C.T. scan uses X-rays from many angles and a computer to make detailed cross-sectional pictures of the body. The C.T. scan is better for treatment because it shows the exact position, size and shape of a disease or injury inside the body, which an ordinary X-ray cannot show clearly.

Q13 Highlight the importance and uses of steel mill.

The steel mill is important because steel is needed for construction, machines, vehicles, ships, tools and many other articles of daily use. Pakistan Steel Mills produces iron and steel from iron ore, which makes the country self-sufficient in steel, provides employment to thousands of workers, and supports other industries like engineering, construction and transport.

Q14 Write a note on pharmaceutical industry.

The pharmaceutical industry makes medicines for the treatment of diseases. In Pakistan it produces tablets, capsules, syrups and injections, and it also makes radio pharmaceuticals for medical use. This industry is important because it provides life-saving drugs at home, reduces dependence on imported medicines, and provides employment to chemists, pharmacists and other workers.

Q15 What are synthetic fibres?

Synthetic fibres are man-made fibres produced by chemical processes from petroleum products, unlike natural fibres obtained

from plants and animals. Examples are nylon, polyester and rayon. They are strong, durable and easy to wash, and they are used for making clothes, ropes, carpets and many other articles.

Q16 What do you know about important sections of textile industry?

The important sections of the textile industry are spinning, in which yarn is made from cotton; weaving, in which cloth is made from yarn; dyeing and printing, in which cloth is coloured and designed; and stitching, in which garments are made. Pakistan's textile industry is very important because it earns a large amount of foreign exchange and provides employment to millions of workers.

Q17 Write a note on leather industry.

The leather industry makes leather from the skins of animals by tanning, and then makes shoes, bags, jackets, gloves and other articles from it. Pakistan's leather industry is important because it earns foreign exchange through exports and provides employment. Its centres are in Lahore, Karachi and Sialkot, where sports goods and leather garments are also made.

Q18 Describe sugar processing in detail.

Sugar processing starts with sugarcane, which is crushed in mills to obtain juice. The juice is cleaned, heated and concentrated by evaporation, and then sugar crystals are formed by cooling. The crystals are separated by centrifuge machines, dried and packed as sugar. The leftover molasses is used for making alcohol and other products, and the bagasse is used as fuel in the mills.

Chapter 11: Space and Nuclear Programme of Pakistan

Exercise 11 - Q1 Fill in the blanks

Q1 (i) Space era started with the launch of _____ Russian artificial satellite.

Sputnik-1

Q1 (ii) By the help of _____ satellites, lot of facilities has been made available in television, telephone and radio communication.

Communication

Q1 (iii) _____ institute looks after the space programme of Pakistan.

SUPARCO

Q1 (iv) The name of first artificial satellite of Pakistan is _____.

Badar-1

Q1 (v) In 1972 the first nuclear power plant was established in _____.

Karachi

Exercise 11 - Q2 True / False

Q2 (i) With the help of space stations, the natural resources hidden in the Earth and the planets of the solar system are explored. — True

Q2 (ii) The man landed on the moon on 20th July 1969. — True

Q2 (iii) The total production capacity of Pakistan's first nuclear power plant is 300 mega watt. — False

Q2 (iv) Cancer can be cured by nuclear radiations. — True

Q2 (v) Skua is the first space rocket of Pakistan. — False

Exercise 11 - Q3 Multiple Choice

Q3 (i) The country, which launched its artificial satellite into space for the first time is

(a) America

(b) France

✓ (c) Russia

(d) Pakistan

Q3 (ii) SUPARCO came into being

(a) in 1956

✓ (b) in 1961

(c) in 1973

(d) in 1990

Q3 (iii) The name of Pakistan's first artificial satellite is

✓ (a) Badar-1

(b) Rahbar

(c) Sputink-1

(d) Skua

Q3 (iv) The total production capacity of Pakistan's first nuclear power plant is

(a) 10 mega watt

✓ (b) 137 mega watt

(c) 300 mega watt

(d) 400 mega watt

Exercise 11 - Q4 Brief Answers

Q4 (i) Which space probes collected the samples of soil and rocks from the surface of Mars?

The Viking space probes collected soil and rock samples from the surface of Mars and tested them for signs of life. Their landers scooped up Martian soil, analysed its chemicals and sent the results and pictures back to the earth.

Q4 (ii) What is meant by Hubble telescope?

The Hubble telescope is a space telescope launched into space on 24 April 1990 with the help of the space shuttle Discovery. The size of its mirror is 2.5 metres and its weight is 11 tonnes. It observes stars, galaxies and planets from above the earth's atmosphere without disturbance.

Q4 (iii) Write down some advantages of artificial satellites.

Some advantages of artificial satellites are: they facilitate television, telephone and radio communication to far off places; they help in weather prediction and warning about cyclones; they help in exploration of natural resources hidden in the earth; and they help scientists study stars, planets, galaxies and cosmic rays.

Q4 (iv) What are the basic objectives of SUPARCO?

The basic objectives of SUPARCO include space research, study of the ozone layer, study of pollution of the atmosphere, astronomy, study of radio waves, exploration of natural mineral resources through geographic information technology, establishment of ground stations, and launching of space rockets and satellites into space for various purposes.

Q4 (v) Where are the nuclear power plants of Pakistan located? What

is their total production capacity?

Pakistan's nuclear power plants are located at Karachi and Chashma. The first nuclear power plant was established at Karachi in 1972 with a total production capacity of 137 megawatts, and later plants were set up at Chashma to increase the production of nuclear electricity.

Exercise 11 - Q5 to Q7 Detailed Questions

Q5 Write a brief note on the importance of space programme. Also describe the utility of artificial satellites.

The space programme is important because information collected by space research is very beneficial for the scientific development of any country. Artificial satellites are useful for television, telephone and radio communication, weather forecasting, exploration of mineral resources, study of the ozone layer and pollution, and scientific study of the universe. Space stations help in studying the natural resources hidden in the earth and the planets of the solar system.

Q6 Write a comprehensive note on the space programme of Pakistan.

Pakistan's space programme is looked after by SUPARCO, the Space and Upper Atmosphere Research Commission, established in 1961 in collaboration with the Pakistan Atomic Energy Commission, with its headquarters at Karachi. Pakistan launched its first rocket, Rahber, into space on 7 June 1962, and by now more than 200 rockets have been launched, hovering from 20 to 550 kilometres. In 1973 three Skua rockets were sent into space to measure air pressure and temperature. In July 1990 Pakistan launched its indigenous artificial satellite Badar-1, and SUPARCO established earth stations at Karachi,

Lahore and Rawat for getting information from satellites.

Q7 What are the services of Pakistan Atomic Energy Commission for the achievement of nuclear energy for peaceful purposes?

The Pakistan Atomic Energy Commission was established in 1956 for the use of nuclear energy for peaceful purposes. Its services are: production of electricity through nuclear power plants at Karachi and Chashma; use of radioisotopes and radiopharmaceuticals in medicine, agriculture and industry; research in nuclear science at PINSTECH, established in 1965; and preparation of special nuclear materials and sophisticated instruments on a limited scale for the needs of the country.